Charge and the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain charge interactions and trace an electric car's energy transfers.

SUCCESS CRITERIA - I can...

1. Predict attraction or repulsion from charge signs.
2. Distinguish charge from energy.
3. Trace energy from a battery to moving wheels.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An ideal blackbody has peak wavelength 966.7 nm. Use λ_{max} $T=2.90\times10^{6}$ nm·K. Estimate temperature.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $E_{electrical} = E_{mechanical} + E_{thermal}$ + other transfers. Energy accounting for a motor with no net energy storage. A battery transfers energy; charge already exists in the wires.

Worked reasoning: Rubbing can transfer electrons between materials. If one isolated object gains charge -3 units and another loses the same electrons, their charge changes add to zero.

Your turn: Draw before-and-after charge accounts for both objects. Distinguish transfer from creation of charge.



Label charges and separation.

Object A Object B

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw force arrows for two positive charges, then for a positive and a negative charge.



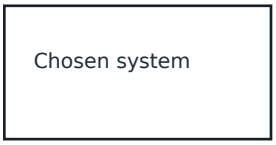
Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

During a model-car test, the battery transfers 120 J and the car gains 30 J of translational kinetic energy. Its height stays fixed. Complete the energy account, find energy in other forms and explain why the circuit's charge is not used up.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says the motor uses up the current so less returns to the battery. Explain the mistake for a steady series circuit.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Three instantaneous displacements are +4 mm, -6 mm and -8 mm. Find their sum and explain why one displacement value does not determine total wave energy.
Show the signed contributions and their sum.

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

07 Exit check

How can an initially neutral object be attracted to a charged object without having a net charge?



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An initially neutral pair separates with charges +8 nC and -8 nC. One then shares all its net charge equally with an identical initially neutral conductor. Find the three final charges.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3